

Library Management System

```
Microsoft Windows [Version 10.0.26200.8524]
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C:\Users\sanvi>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 15
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE library_management;
Query OK, 1 row affected (0.09 sec)

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| library_management |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.09 sec)

mysql> USE library_management;
Database changed
mysql> SELECT DATABASE();
+-----+
| DATABASE() |
+-----+
| library_management |
+-----+
1 row in set (0.00 sec)

mysql> CREATE TABLE Author(
  -> author_id INT PRIMARY KEY,
  -> name VARCHAR(100),
  -> country VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.07 sec)

mysql> DESC Author;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| author_id | int | NO | PRI | NULL | |
| name | varchar(100) | YES | | NULL | |
| country | varchar(50) | YES | | NULL | |
+-----+
3 rows in set (0.02 sec)

mysql> CREATE TABLE Member(
  -> member_id INT PRIMARY KEY,
  -> name VARCHAR(100),
  -> phone VARCHAR(15),
  -> membership_date DATE
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> DESC Member;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| member_id | int | NO | PRI | NULL | |
| name | varchar(100) | YES | | NULL | |
| phone | varchar(15) | YES | | NULL | |
| membership_date | date | YES | | NULL | |
+-----+
4 rows in set (0.00 sec)

mysql> CREATE TABLE Book(
  -> book_id INT PRIMARY KEY,
  -> title VARCHAR(100),
  -> author_id INT,
  -> category VARCHAR(50),
  -> price DECIMAL(10,2),
  -> availability VARCHAR(10),
  ->
  -> FOREIGN KEY(author_id)
  -> REFERENCES Author(author_id)
  -> );
Query OK, 0 rows affected (0.07 sec)

mysql> DESC Book;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| book_id | int | NO | PRI | NULL | |
| title | varchar(100) | YES | | NULL | |
| author_id | int | YES | | NULL | |
| category | varchar(50) | YES | | NULL | |
| price | decimal(10,2) | YES | | NULL | |
| availability | varchar(10) | YES | | NULL | |
+-----+
6 rows in set (0.00 sec)
```

```
Command Prompt - mysql -u +
mysql> DESC Book;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| book_id | int | NO | PRI | NULL | |
| title | varchar(100) | YES | | NULL | |
| author_id | int | YES | MUL | NULL | |
| category | varchar(50) | YES | | NULL | |
| price | decimal(10,2) | YES | | NULL | |
| availability | varchar(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> CREATE TABLE Issue(
-> issue_id INT PRIMARY KEY,
-> member_id INT,
-> book_id INT,
-> issue_date DATE,
-> return_date DATE,
->
-> FOREIGN KEY(member_id)
-> REFERENCES Member(member_id),
->
-> FOREIGN KEY(book_id)
-> REFERENCES Book(book_id)
-> );
Query OK, 0 rows affected (0.09 sec)

mysql> DESC Issue;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| issue_id | int | NO | PRI | NULL | |
| member_id | int | YES | MUL | NULL | |
| book_id | int | YES | MUL | NULL | |
| issue_date | date | YES | | NULL | |
| return_date | date | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> CREATE TABLE Fine(
-> fine_id INT PRIMARY KEY,
-> member_id INT,
-> amount DECIMAL(10,2),
-> paid_status VARCHAR(20),
->
-> FOREIGN KEY(member_id)
```

```
Command Prompt - mysql -u +
-> REFERENCES Member(member_id)
-> );
Query OK, 0 rows affected (0.08 sec)

mysql> DESC Fine;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| fine_id | int | NO | PRI | NULL | |
| member_id | int | YES | MUL | NULL | |
| amount | decimal(10,2) | YES | | NULL | |
| paid_status | varchar(20) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> CREATE TABLE Staff(
-> staff_id INT PRIMARY KEY,
-> name VARCHAR(100),
-> role VARCHAR(50)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> DESC Staff;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| staff_id | int | NO | PRI | NULL | |
| name | varchar(100) | YES | | NULL | |
| role | varchar(50) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> INSERT INTO Author VALUES
-> (1,'Chetan Bhagat','India'),
-> (2,'J K Rowling','UK'),
-> (3,'Dan Brown','USA');
Query OK, 3 rows affected (0.03 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Author;
+-----+-----+-----+
| author_id | name | country |
+-----+-----+-----+
| 1 | Chetan Bhagat | India |
| 2 | J K Rowling | UK |
| 3 | Dan Brown | USA |
+-----+-----+-----+
```

```
Command Prompt - mysql -u x + v
3 rows in set (0.00 sec)

mysql> INSERT INTO Member VALUES
-> (101,'Sanvi','9876543210','2025-01-01'),
-> (102,'Rahul','9876543211','2025-01-02'),
-> (103,'Ananya','9876543212','2025-01-03');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Member;
+-----+-----+-----+-----+
| member_id | name   | phone   | membership_date |
+-----+-----+-----+-----+
| 101       | Sanvi  | 9876543210 | 2025-01-01      |
| 102       | Rahul  | 9876543211 | 2025-01-02      |
| 103       | Ananya | 9876543212 | 2025-01-03      |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> INSERT INTO Book VALUES
-> (201,'Five Point Someone',1,'Novel',300,'Available'),
-> (202,'Harry Potter',2,'Fantasy',600,'Available'),
-> (203,'Da Vinci Code',3,'Mystery',450,'Available');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Book;
+-----+-----+-----+-----+-----+-----+
| book_id | title           | author_id | category | price | availability |
+-----+-----+-----+-----+-----+-----+
| 201     | Five Point Someone | 1         | Novel    | 300.00 | Available    |
| 202     | Harry Potter       | 2         | Fantasy  | 600.00 | Available    |
| 203     | Da Vinci Code      | 3         | Mystery  | 450.00 | Available    |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> INSERT INTO Issue VALUES
-> (1,101,201,'2025-05-01','2025-05-10'),
-> (2,102,202,'2025-05-02','2025-05-12');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Issue;
+-----+-----+-----+-----+-----+
| issue_id | member_id | book_id | issue_date | return_date |
+-----+-----+-----+-----+-----+
| 1        | 101       | 201     | 2025-05-01 | 2025-05-10  |
| 2        | 102       | 202     | 2025-05-02 | 2025-05-12  |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
Command Prompt - mysql -u x + v

mysql> SELECT * FROM Issue;
+-----+-----+-----+-----+-----+
| issue_id | member_id | book_id | issue_date | return_date |
+-----+-----+-----+-----+-----+
| 1        | 101       | 201     | 2025-05-01 | 2025-05-10  |
| 2        | 102       | 202     | 2025-05-02 | 2025-05-12  |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> INSERT INTO Fine VALUES
-> (1,101,50,'No'),
-> (2,102,0,'Yes');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Fine;
+-----+-----+-----+-----+
| fine_id | member_id | amount | paid_status |
+-----+-----+-----+-----+
| 1       | 101       | 50.00  | No          |
| 2       | 102       | 0.00   | Yes         |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> INSERT INTO Staff VALUES
-> (1,'Ramesh','Librarian'),
-> (2,'Suresh','Assistant');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Staff;
+-----+-----+-----+
| staff_id | name   | role   |
+-----+-----+-----+
| 1        | Ramesh | Librarian |
| 2        | Suresh | Assistant |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_library_management |
+-----+
| author                       |
| book                         |
| fine                         |
| issue                        |
+-----+
```

```
Command Prompt - mysql -u x + v
mysql> SHOW TABLES;
+-----+
| Tables_in_library_management |
+-----+
| author |
| book |
| fine |
| issue |
| member |
| staff |
+-----+
6 rows in set (0.01 sec)

mysql> DESC Book;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| book_id | int | NO | PRI | NULL | |
| title | varchar(100) | YES | | NULL | |
| author_id | int | YES | MUL | NULL | |
| category | varchar(50) | YES | | NULL | |
| price | decimal(10,2) | YES | | NULL | |
| availability | varchar(10) | YES | | NULL | |
+-----+
6 rows in set (0.00 sec)

mysql> SELECT * FROM Author;
+-----+
| author_id | name | country |
+-----+
| 1 | Chetan Bhagat | India |
| 2 | J K Rowling | UK |
| 3 | Dan Brown | USA |
+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Member;
+-----+
| member_id | name | phone | membership_date |
+-----+
| 101 | Sanvi | 9876543210 | 2025-01-01 |
| 102 | Rahul | 9876543211 | 2025-01-02 |
| 103 | Ananya | 9876543212 | 2025-01-03 |
+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Book;
```

```
Command Prompt - mysql -u x + v
mysql> SELECT * FROM Book;
+-----+
| book_id | title | author_id | category | price | availability |
+-----+
| 201 | Five Point Someone | 1 | Novel | 300.00 | Available |
| 202 | Harry Potter | 2 | Fantasy | 600.00 | Available |
| 203 | Da Vinci Code | 3 | Mystery | 450.00 | Available |
+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Issue;
+-----+
| issue_id | member_id | book_id | issue_date | return_date |
+-----+
| 1 | 101 | 201 | 2025-05-01 | 2025-05-10 |
| 2 | 102 | 202 | 2025-05-02 | 2025-05-12 |
+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Fine;
+-----+
| fine_id | member_id | amount | paid_status |
+-----+
| 1 | 101 | 50.00 | No |
| 2 | 102 | 0.00 | Yes |
+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Staff;
+-----+
| staff_id | name | role |
+-----+
| 1 | Ramesh | Librarian |
| 2 | Suresh | Assistant |
+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Book;
+-----+
| book_id | title | author_id | category | price | availability |
+-----+
| 201 | Five Point Someone | 1 | Novel | 300.00 | Available |
| 202 | Harry Potter | 2 | Fantasy | 600.00 | Available |
| 203 | Da Vinci Code | 3 | Mystery | 450.00 | Available |
+-----+
3 rows in set (0.00 sec)
```

```
Command Prompt - mysql -u x + v
mysql> SELECT *
-> FROM Book
-> WHERE author_id=1;
+-----+-----+-----+-----+-----+-----+
| book_id | title           | author_id | category | price | availability |
+-----+-----+-----+-----+-----+-----+
| 201     | Five Point Someone | 1         | Novel    | 300.00 | Available    |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT *
-> FROM Issue i
-> INNER JOIN Member m
-> ON i.member_id=m.member_id;
+-----+-----+-----+-----+-----+-----+-----+-----+
| issue_id | member_id | book_id | issue_date | return_date | member_id | name | phone | membership_date |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1        | 101       | 201     | 2025-05-01 | 2025-05-10 | 101      | Sanvi | 9876543210 | 2025-01-01      |
| 2        | 102       | 202     | 2025-05-02 | 2025-05-12 | 102      | Rahul | 9876543211 | 2025-01-02      |
+-----+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT *
-> FROM Issue i
-> JOIN Member m
-> ON i.member_id=m.member_id
-> JOIN Book b
-> ON i.book_id=b.book_id;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| issue_id | member_id | book_id | issue_date | return_date | member_id | name | phone | membership_date | book_id | title           | author_id |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1        | 101       | 201     | 2025-05-01 | 2025-05-10 | 101      | Sanvi | 9876543210 | 2025-01-01      | 201     | Five Point Someone | 1         |
| 2        | 102       | 202     | 2025-05-02 | 2025-05-12 | 102      | Rahul | 9876543211 | 2025-01-02      | 202     | Harry Potter      | 2         |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT category,
-> COUNT(*) AS TotalBooks
-> FROM Book
-> GROUP BY category;
+-----+-----+
| category | TotalBooks |
+-----+-----+
| Novel    | 1          |
| Fantasy  | 1          |
| Mystery  | 1          |
+-----+-----+
3 rows in set (0.01 sec)
```

```
Command Prompt - mysql -u x + v
-> GROUP BY category;
+-----+-----+
| category | TotalBooks |
+-----+-----+
| Novel    | 1          |
| Fantasy  | 1          |
| Mystery  | 1          |
+-----+-----+
3 rows in set (0.01 sec)

mysql> SELECT category,
-> COUNT(*) AS TotalBooks
-> FROM Book
-> GROUP BY category
-> HAVING COUNT(*)>50;
Empty set (0.00 sec)

mysql> SELECT *
-> FROM Book
-> WHERE price >
-> (
-> SELECT AVG(price)
-> FROM Book
-> );
+-----+-----+-----+-----+-----+-----+
| book_id | title           | author_id | category | price | availability |
+-----+-----+-----+-----+-----+-----+
| 202     | Harry Potter    | 2         | Fantasy  | 600.00 | Available    |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT m1.*
-> FROM Member m1
-> WHERE
-> (
-> SELECT COUNT(*)
-> FROM Issue i
-> WHERE i.member_id=m1.member_id
-> )
-> >
-> (
-> SELECT COUNT(*)
-> FROM Issue i
-> WHERE i.member_id=101
-> );
Empty set (0.00 sec)
```

```

mysql> SELECT *
-> FROM Book b
-> LEFT JOIN Issue i
-> ON b.book_id=i.book_id;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| book_id | title           | author_id | category | price | availability | issue_id | member_id | book_id | issue_date | return_date |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 201     | Five Point Someone | 1         | Novel    | 300.00 | Available    | 1        | 101       | 201     | 2025-05-01 | 2025-05-10 |
| 202     | Harry Potter       | 2         | Fantasy  | 600.00 | Available    | 2        | 102       | 202     | 2025-05-02 | 2025-05-12 |
| 203     | Da Vinci Code      | 3         | Mystery  | 450.00 | Available    | NULL     | NULL      | NULL    | NULL       | NULL        |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT *
-> FROM Book b
-> WHERE NOT EXISTS
-> (
-> SELECT *
-> FROM Issue i
-> WHERE b.book_id=i.book_id
-> );
+-----+-----+-----+-----+-----+-----+
| book_id | title           | author_id | category | price | availability |
+-----+-----+-----+-----+-----+-----+
| 203     | Da Vinci Code   | 3         | Mystery  | 450.00 | Available    |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_library_management |
+-----+
| author                       |
| book                         |
| fine                         |
| issue                        |
| member                       |
| staff                        |
+-----+
6 rows in set (0.00 sec)

mysql> DESC Book;
+-----+-----+-----+-----+-----+-----+
| Field | Type           | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| book_id | int            | NO   | PRI | NULL    |       |
| title   | varchar(100)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+

```

```

mysql> DESC Book;
+-----+-----+-----+-----+-----+-----+
| Field | Type           | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| book_id | int            | NO   | PRI | NULL    |       |
| title   | varchar(100)   | YES  |     | NULL    |       |
| author_id | int          | YES  | MUL | NULL    |       |
| category | varchar(50)    | YES  |     | NULL    |       |
| price   | decimal(10,2)  | YES  |     | NULL    |       |
| availability | varchar(10)  | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> SELECT * FROM Author;
+-----+-----+-----+
| author_id | name           | country |
+-----+-----+-----+
| 1         | Chetan Bhagat | India   |
| 2         | J K Rowling    | UK      |
| 3         | Dan Brown      | USA     |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Member;
+-----+-----+-----+-----+
| member_id | name           | phone           | membership_date |
+-----+-----+-----+-----+
| 101       | Sanvi          | 9876543210      | 2025-01-01      |
| 102       | Rahul          | 9876543211      | 2025-01-02      |
| 103       | Ananya         | 9876543212      | 2025-01-03      |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Book;
+-----+-----+-----+-----+-----+-----+
| book_id | title           | author_id | category | price | availability |
+-----+-----+-----+-----+-----+-----+
| 201     | Five Point Someone | 1         | Novel    | 300.00 | Available    |
| 202     | Harry Potter       | 2         | Fantasy  | 600.00 | Available    |
| 203     | Da Vinci Code      | 3         | Mystery  | 450.00 | Available    |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Issue;
+-----+-----+-----+-----+-----+
| issue_id | member_id | book_id | issue_date | return_date |
+-----+-----+-----+-----+-----+

```

```
Command Prompt - mysql -u x + v

mysql> SELECT * FROM Issue;
+-----+-----+-----+-----+-----+
| issue_id | member_id | book_id | issue_date | return_date |
+-----+-----+-----+-----+-----+
| 1 | 101 | 201 | 2025-05-01 | 2025-05-10 |
| 2 | 102 | 202 | 2025-05-02 | 2025-05-12 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Fine;
+-----+-----+-----+-----+
| fine_id | member_id | amount | paid_status |
+-----+-----+-----+-----+
| 1 | 101 | 50.00 | No |
| 2 | 102 | 0.00 | Yes |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Staff;
+-----+-----+-----+
| staff_id | name | role |
+-----+-----+-----+
| 1 | Ramesh | Librarian |
| 2 | Suresh | Assistant |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE IssueBook
-> (
-> IN p_issueid INT,
-> IN p_memberid INT,
-> IN p_bookid INT,
-> IN p_issuedate DATE,
-> IN p_returndate DATE
-> )
->
-> BEGIN
->
-> INSERT INTO Issue
-> VALUES
-> (
-> p_issueid,
-> p_memberid,
-> p_bookid,
-> )
->
-> END
->
Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER ;
mysql> CALL IssueBook
-> (
-> 3,
-> 103,
-> 203,
-> '2025-05-20',
-> '2025-05-30'
-> );
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Issue;
+-----+-----+-----+-----+-----+
| issue_id | member_id | book_id | issue_date | return_date |
+-----+-----+-----+-----+-----+
| 1 | 101 | 201 | 2025-05-01 | 2025-05-10 |
| 2 | 102 | 202 | 2025-05-02 | 2025-05-12 |
| 3 | 103 | 203 | 2025-05-20 | 2025-05-30 |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE CalculateFine
-> (
-> IN days_late INT
-> )
->
-> BEGIN
->
-> SELECT days_late*5
-> AS FineAmount;
->
-> END
->
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL CalculateFine(10);
```

```
Command Prompt - mysql -u x + v

-> p_issuedate,
-> p_returndate
-> );
->
-> END $$
Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER ;
mysql> CALL IssueBook
-> (
-> 3,
-> 103,
-> 203,
-> '2025-05-20',
-> '2025-05-30'
-> );
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Issue;
+-----+-----+-----+-----+-----+
| issue_id | member_id | book_id | issue_date | return_date |
+-----+-----+-----+-----+-----+
| 1 | 101 | 201 | 2025-05-01 | 2025-05-10 |
| 2 | 102 | 202 | 2025-05-02 | 2025-05-12 |
| 3 | 103 | 203 | 2025-05-20 | 2025-05-30 |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE CalculateFine
-> (
-> IN days_late INT
-> )
->
-> BEGIN
->
-> SELECT days_late*5
-> AS FineAmount;
->
-> END
->
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL CalculateFine(10);
```

```
Command Prompt - mysql -u x + v
mysql> CALL CalculateFine(10);
+-----+
| FineAmount |
+-----+
|          50 |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER $$
mysql> CREATE TRIGGER trg_book_issue
--
-- AFTER INSERT
-- ON Issue
--
-- FOR EACH ROW
--
-- BEGIN
--
-- UPDATE Book
--
-- SET availability='Issued'
--
-- WHERE book_id=NEW.book_id;
--
-- END $$
Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER ;
mysql> INSERT INTO Issue
-- VALUES
--
-- (4,101,203,'2025-05-15','2025-05-25');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Book;
+-----+-----+-----+-----+-----+-----+
| book_id | title           | author_id | category | price  | availability |
+-----+-----+-----+-----+-----+-----+
|      201 | Five Point Someone |          1 | Novel    | 300.00 | Available    |
|      202 | Harry Potter       |          2 | Fantasy  | 600.00 | Available    |
|      203 | Da Vinci Code      |          3 | Mystery  | 450.00 | Issued       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
Command Prompt - mysql -u x + v
mysql> ALTER TABLE Issue
--> ADD due_date DATE;
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DELIMITER $$
mysql> CREATE TRIGGER trg_fine
--
-- AFTER UPDATE
-- ON Issue
--
-- FOR EACH ROW
--
-- BEGIN
--
-- IF NEW.return_date > NEW.due_date THEN
--
-- INSERT INTO Fine
-- VALUES
-- (
--
-- 3,
--
-- NEW.member_id,
--
-- DATEDIFF
-- (
--
-- NEW.return_date,
--
-- NEW.due_date
--
-- )+5,
--
-- 'No'
--
-- );
--
-- END IF;
--
-- END $$
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> SHOW TABLES;
+-----+
| Tables_in_library_management |
+-----+
| author                       |
| book                         |
| fine                         |
| issue                        |
| member                       |
+-----+
```



```
Command Prompt - mysql -u x + v
mysql> SHOW TABLES;
+-----+
| Tables_in_library_management |
+-----+
| author |
| book |
| fine |
| issue |
| member |
| staff |
+-----+
6 rows in set (0.00 sec)

mysql> DESC Book;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| book_id | int | NO | PRI | NULL | |
| title | varchar(100) | YES | | NULL | |
| author_id | int | YES | MUL | NULL | |
| category | varchar(50) | YES | | NULL | |
| price | decimal(10,2) | YES | | NULL | |
| availability | varchar(10) | YES | | NULL | |
+-----+
6 rows in set (0.00 sec)

mysql> SELECT * FROM Author;
+-----+
| author_id | name | country |
+-----+
| 1 | Chetan Bhagat | India |
| 2 | J K Rowling | UK |
| 3 | Dan Brown | USA |
+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Member;
+-----+
| member_id | name | phone | membership_date |
+-----+
| 101 | Sanvi | 9876543210 | 2025-01-01 |
| 102 | Rahul | 9876543211 | 2025-01-02 |
| 103 | Ananya | 9876543212 | 2025-01-03 |
+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Book;
```

```
Command Prompt - mysql -u x + v
mysql> SELECT * FROM Book;
+-----+
| book_id | title | author_id | category | price | availability |
+-----+
| 201 | Five Point Someone | 1 | Novel | 300.00 | Available |
| 202 | Harry Potter | 2 | Fantasy | 600.00 | Available |
| 203 | Da Vinci Code | 3 | Mystery | 450.00 | Issued |
+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Issue;
+-----+
| issue_id | member_id | book_id | issue_date | return_date | due_date |
+-----+
| 1 | 101 | 201 | 2025-05-01 | 2025-05-10 | NULL |
| 2 | 102 | 202 | 2025-05-02 | 2025-05-12 | NULL |
| 3 | 103 | 203 | 2025-05-20 | 2025-05-30 | NULL |
| 4 | 101 | 203 | 2025-05-15 | 2025-05-25 | NULL |
+-----+
4 rows in set (0.00 sec)

mysql> SELECT * FROM Fine;
+-----+
| fine_id | member_id | amount | paid_status |
+-----+
| 1 | 101 | 50.00 | No |
| 2 | 102 | 0.00 | Yes |
+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Staff;
+-----+
| staff_id | name | role |
+-----+
| 1 | Ramesh | Librarian |
| 2 | Suresh | Assistant |
+-----+
2 rows in set (0.00 sec)

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| library_management |
| mysql |
+-----+
```

```
Command Prompt - mysql -u + -
3 rows in set (0.00 sec)

mysql> SELECT * FROM Issue;
+-----+-----+-----+-----+-----+-----+
| issue_id | member_id | book_id | issue_date | return_date | due_date |
+-----+-----+-----+-----+-----+-----+
| 1 | 101 | 201 | 2025-05-01 | 2025-05-10 | NULL |
| 2 | 102 | 202 | 2025-05-02 | 2025-05-12 | NULL |
| 3 | 103 | 203 | 2025-05-20 | 2025-05-30 | NULL |
| 4 | 101 | 203 | 2025-05-15 | 2025-05-25 | NULL |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT * FROM Fine;
+-----+-----+-----+-----+
| fine_id | member_id | amount | paid_status |
+-----+-----+-----+-----+
| 1 | 101 | 50.00 | No |
| 2 | 102 | 0.00 | Yes |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Staff;
+-----+-----+-----+
| staff_id | name | role |
+-----+-----+-----+
| 1 | Ramesh | Librarian |
| 2 | Suresh | Assistant |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| library_management |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.00 sec)

mysql> |
```